

studentai

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Chapter 1

Hierarchical Index

1.1 Class Hierarchy

This inheritance list is sorted roughly, but not completely, alphabetically:

Human	7
Studentas	8

Chapter 2

Class Index

2.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

Human		7
Studentas		8

Chapter 3

File Index

3.1 File List

Here is a list of all files with brief descriptions:

Include/human.h	13
Include/my_functions.h	13
Include/my_library.h	17
Include/student.h	18
src/additional_functions.cpp	20
src/file_functions.cpp	20
src/main.cpp	21
src/program_choice.cpp	21
src/random_generating.cpp	22
src/student.cpp	23
src/student_functions.cpp	23
src/testavimas.cpp	24

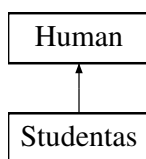
Chapter 4

Class Documentation

4.1 Human Class Reference

```
#include <human.h>
```

Inheritance diagram for Human:



Public Member Functions

- `Human` (const string &`vardas`, const string &`pavarde`)
- string `getVardas` () const
- string `getPavarde` () const
- virtual void `unusedFunction` () const =0
- virtual `~Human` ()

Protected Attributes

- string `vardas`
- string `pavarde`

4.1.1 Constructor & Destructor Documentation

4.1.1.1 Human()

```
Human::Human (  
    const string & vardas,  
    const string & pavarde) [inline]
```

4.1.1.2 ~Human()

```
virtual Human::~~Human () [inline], [virtual]
```

4.1.2 Member Function Documentation

4.1.2.1 getPavarde()

```
string Human::getPavarde () const [inline]
```

4.1.2.2 getVardas()

```
string Human::getVardas () const [inline]
```

4.1.2.3 unusedFunction()

```
virtual void Human::unusedFunction () const [pure virtual]
```

Implemented in [Studentas](#).

4.1.3 Member Data Documentation

4.1.3.1 pavarde

```
string Human::pavarde [protected]
```

4.1.3.2 vardas

```
string Human::vardas [protected]
```

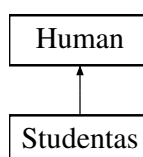
The documentation for this class was generated from the following file:

- Include/[human.h](#)

4.2 Studentas Class Reference

```
#include <student.h>
```

Inheritance diagram for Studentas:



Public Member Functions

- void `unusedFunction` () const override
- `Studentas` ()
- `Studentas` (const string &`vardas`, const string &`pavarde`, const vector< float > &pazymiai, int egzaminoRezultatas)
- `~Studentas` ()
- `Studentas` (const `Studentas` &other)
- `Studentas` (`Studentas` &&other) noexcept
- `Studentas` & `operator=` (const `Studentas` &other)
- `Studentas` & `operator=` (`Studentas` &&other)
- void `setVardas` (const string &v)
- void `setPavarde` (const string &p)
- void `setGrades` (const vector< float > &g)
- void `setExam` (int exam)
- vector< float > `getPazymiai` () const
- int `getEgzaminoRezultatas` () const
- float `getGalutinisVid` () const
- float `getGalutinisMed` () const
- float `calculateMean` ()
- float `calculateMedian` ()
- void `calculate_everything` ()

Public Member Functions inherited from `Human`

- `Human` (const string &`vardas`, const string &`pavarde`)
- string `getVardas` () const
- string `getPavarde` () const
- virtual `~Human` ()

Static Public Member Functions

- static void `appendingContainerViaFile` (string file_name, vector< `Studentas` > &student_list)
- static vector< `Studentas` > `read_student_records` (int ndCount, std::istream &iss)
- static void `print_to_file` (vector< `Studentas` > list_of_students, string file_name)

Friends

- ostream & `operator<<` (ostream &out, const `Studentas` &s)
- std::istream & `operator>>` (std::istream &in, `Studentas` &s)

Additional Inherited Members

Protected Attributes inherited from `Human`

- string `vardas`
- string `pavarde`

4.2.1 Constructor & Destructor Documentation

4.2.1.1 Studentas() [1/4]

```
Studentas::Studentas () [inline]
```

4.2.1.2 Studentas() [2/4]

```
Studentas::Studentas (  
    const string & vardas,  
    const string & pavarde,  
    const vector< float > & pazymiai,  
    int egzaminoRezultatas) [inline]
```

4.2.1.3 ~Studentas()

```
Studentas::~~Studentas () [inline]
```

4.2.1.4 Studentas() [3/4]

```
Studentas::Studentas (  
    const Studentas & other)
```

4.2.1.5 Studentas() [4/4]

```
Studentas::Studentas (  
    Studentas && other) [noexcept]
```

4.2.2 Member Function Documentation

4.2.2.1 appendingContainerViaFile()

```
void Studentas::appendingContainerViaFile (  
    string file_name,  
    vector< Studentas > & student_list) [static]
```

4.2.2.2 calculate_everything()

```
void Studentas::calculate_everything ()
```

4.2.2.3 calculateMean()

```
float Studentas::calculateMean ()
```

4.2.2.4 calculateMedian()

```
float Studentas::calculateMedian ()
```

4.2.2.5 getEgzaminoRezultatas()

```
int Studentas::getEgzaminoRezultatas () const [inline]
```

4.2.2.6 getGalutinisMed()

```
float Studentas::getGalutinisMed () const [inline]
```

4.2.2.7 getGalutinisVid()

```
float Studentas::getGalutinisVid () const [inline]
```

4.2.2.8 getPazymiai()

```
vector< float > Studentas::getPazymiai () const [inline]
```

4.2.2.9 operator=() [1/2]

```
Studentas & Studentas::operator= (  
    const Studentas & other)
```

4.2.2.10 operator=() [2/2]

```
Studentas & Studentas::operator= (  
    Studentas && other)
```

4.2.2.11 print_to_file()

```
void Studentas::print_to_file (  
    vector< Studentas > list_of_students,  
    string file_name) [static]
```

4.2.2.12 read_student_records()

```
vector< Studentas > Studentas::read_student_records (  
    int ndCount,  
    std::istream & iss) [static]
```

4.2.2.13 setExam()

```
void Studentas::setExam (
    int exam) [inline]
```

4.2.2.14 setGrades()

```
void Studentas::setGrades (
    const vector< float > & g) [inline]
```

4.2.2.15 setPavarde()

```
void Studentas::setPavarde (
    const string & p) [inline]
```

4.2.2.16 setVardas()

```
void Studentas::setVardas (
    const string & v) [inline]
```

4.2.2.17 unusedFunction()

```
void Studentas::unusedFunction () const [override], [virtual]
```

Implements [Human](#).

4.2.3 Friends And Related Symbol Documentation

4.2.3.1 operator<<

```
ostream & operator<< (
    ostream & out,
    const Studentas & s) [friend]
```

4.2.3.2 operator>>

```
std::istream & operator>> (
    std::istream & in,
    Studentas & s) [friend]
```

The documentation for this class was generated from the following files:

- Include/[student.h](#)
- src/[student.cpp](#)

Chapter 5

File Documentation

5.1 Include/human.h File Reference

```
#include "my_library.h"
```

Classes

- class [Human](#)

5.2 human.h

[Go to the documentation of this file.](#)

```
00001 #pragma once
00002
00003 #include "my_library.h"
00004
00005 class Human {
00006     protected:
00007         string vardas;
00008         string pavarde;
00009
00010     public:
00011         Human(const string& vardas, const string& pavarde)
00012             : vardas(vardas), pavarde(pavarde)
00013             { }
00014
00015         string getVardas() const { return vardas; }
00016         string getPavarde() const { return pavarde; }
00017
00018         virtual void unusedFunction() const = 0;
00019
00020         // Virtual destructor for proper cleanup
00021         virtual ~Human() {}
00022
00023 };
```

5.3 Include/my_functions.h File Reference

```
#include "my_library.h"
#include "student.h"
```

Functions

- string [create_folder](#) (string folder_name)
file_functions.cpp
- void [generate_files](#) (int student_size)
- void [print_students](#) (vector< [Studentas](#) > &studentas)
student_functions.cpp
- void [insert_student](#) (vector< [Studentas](#) > &student_list, [Studentas](#) &student)
- void [sort_students](#) (vector< [Studentas](#) > &student_list)
- void [sort_students](#) (vector< [Studentas](#) > &student_list, int choice)
- [Studentas](#) [userChoice_2](#) ()
program_choice.cpp
- [Studentas](#) [userChoice_3](#) ()
- int [randomNumber](#) (int a, int b)
- vector< float > [random_grade](#) ()
- string [random_name](#) ()
- string [random_last_name](#) ()
- string [consoleText_userChoice](#) ()
- int [check_the_value](#) (string message, string errorMessage, int minVal, int maxVal)
- void [testing_v12](#) ()
- void [testing_v11](#) ()
- void [strategy_1](#) ()
- void [strategy_2](#) ()
- void [strategy_3](#) ()

5.3.1 Function Documentation

5.3.1.1 check_the_value()

```
int check_the_value (
    string message,
    string errorMessage,
    int minVal,
    int maxVal)
```

5.3.1.2 consoleText_userChoice()

```
string consoleText_userChoice ()
```

5.3.1.3 create_folder()

```
string create_folder (
    string folder_name)
```

[file_functions.cpp](#)

5.3.1.4 generate_files()

```
void generate_files (
    int student_size)
```

5.3.1.5 insert_student()

```
void insert_student (
    vector< Studentas > & student_list,
    Studentas & student)
```

5.3.1.6 print_students()

```
void print_students (
    vector< Studentas > & studentas)
```

[student_functions.cpp](#)

5.3.1.7 random_grade()

```
vector< float > random_grade ()
```

5.3.1.8 random_last_name()

```
string random_last_name ()
```

5.3.1.9 random_name()

```
string random_name ()
```

5.3.1.10 randomNumber()

```
int randomNumber (
    int a,
    int b)
```

5.3.1.11 sort_students() [1/2]

```
void sort_students (
    vector< Studentas > & student_list)
```

5.3.1.12 sort_students() [2/2]

```
void sort_students (
    vector< Studentas > & student_list,
    int choice)
```

5.3.1.13 strategy_1()

```
void strategy_1 ()
```

<— sparta pradedama skaiciuoti cia

5.3.1.14 strategy_2()

```
void strategy_2 ()
```

5.3.1.15 strategy_3()

```
void strategy_3 ()
```

5.3.1.16 testing_v11()

```
void testing_v11 ()
```

5.3.1.17 testing_v12()

```
void testing_v12 ()
```

5.3.1.18 userChoice_2()

```
Studentas userChoice_2 ()
```

[program_choice.cpp](#)

5.3.1.19 userChoice_3()

```
Studentas userChoice_3 ()
```

5.4 my_functions.h

[Go to the documentation of this file.](#)

```

00001 #pragma once
00002
00003 #include "my_library.h"
00004 #include "student.h"
00005
00006
00007 // -----
00008 namespace fs = std::filesystem;
00009 string create_folder(string folder_name);
00010
00011 void generate_files(int student_size);
00012
00013 // -----
00014
00015 void print_students(vector<Studentas> &studentas);
00016 void insert_student(vector<Studentas> &student_list, Studentas &student);
00017
00018 void sort_students(vector<Studentas> &student_list);
00019 void sort_students(vector<Studentas> &student_list, int choice);
00020
00021 // -----
00022 Studentas userChoice_2();
00023 Studentas userChoice_3();
00024
00025 // -----
00026 // random_generating.cpp
00027
00028 int randomNumber(int a, int b);
00029 vector<float> random_grade();
00030
00031 string random_name();
00032 string random_last_name();
00033
00034 // -----
00035 // additional_functions.cpp
00036 string consoleText_userChoice();
00037 int check_the_value(string message, string errorMessage, int minVal, int maxVal); // checks if a
00038 give value is within range./
00039
00040 // -----
00041 // testing.cpp
00042 void testing_vl2();
00043 void testing_vl1();
00044 void strategy_1();
00045 void strategy_2();
00046 void strategy_3();

```

5.5 Include/my_library.h File Reference

```

#include <vector>
#include <iostream>
#include <fstream>
#include <sstream>
#include <string>
#include <iomanip>
#include <algorithm>
#include <iterator>
#include <limits>
#include <numeric>
#include <random>
#include <cmath>
#include <chrono>
#include <filesystem>

```

5.6 my_library.h

[Go to the documentation of this file.](#)

```
00001 #pragma once
00002
00003
00004
00005 // --- Containers ---
00006 #include <vector>
00007 using std::vector;
00008
00009 // --- I/O Streams ---
00010 #include <iostream>
00011 using std::cout;
00012 using std::cin;
00013 using std::cerr;
00014 using std::endl;
00015 using std::ostream;
00016 using std::ios;
00017
00018 #include <fstream>
00019 using std::ifstream;
00020 using std::ofstream;
00021
00022 #include <sstream>
00023 using std::istringstream;
00024 using std::ostringstream;
00025
00026 // --- String and Formatting ---
00027 #include <string>
00028 using std::string;
00029 using std::to_string;
00030
00031 #include <iomanip>
00032 using std::setw;
00033 using std::fixed;
00034 using std::setprecision;
00035 using std::left;
00036
00037 // --- Algorithms and Iterators ---
00038 #include <algorithm>
00039 using std::sort;
00040
00041 #include <iterator>
00042 using std::istream_iterator;
00043
00044 // --- Numeric Utilities ---
00045 #include <limits>
00046 using std::numeric_limits;
00047 using std::streamsize;
00048
00049 #include <numeric>
00050 using std::accumulate;
00051
00052 // --- Random Number Generation ---
00053 #include <random>
00054 using std::random_device;
00055 using std::mt19937;
00056 using std::uniform_int_distribution;
00057
00058 // --- Math ---
00059 #include <cmath>
00060
00061 // --- Time Utilities ---
00062 #include <chrono>
00063 using std::chrono::high_resolution_clock;
00064 using std::chrono::duration;
00065
00066 #include <filesystem>
```

5.7 Include/student.h File Reference

```
#include "human.h"
#include "my_library.h"
```

Classes

- class [Studentas](#)

5.8 student.h

[Go to the documentation of this file.](#)

```

00001 #pragma once
00002
00003 #include "human.h"
00004 #include "my_library.h"
00005
00006 class Studentas : public Human {
00007     private:
00008         vector<float> pazymiai;
00009         int egzaminoRezultatas;
00010
00011         float galutinisVid;
00012         float galutinisMed;
00013
00014     public:
00015
00016         // Override the pure virtual function from Human.
00017         void unusedFunction() const override;
00018
00019         // Default constructor
00020         Studentas()
00021             : Human("", ""), egzaminoRezultatas(0),
00022               galutinisVid(0.0f), galutinisMed(0.0f) {}
00023
00024         // Parameterized constructor
00025         Studentas(const string& vardas, const string& pavarde, const vector<float>& pazymiai, int
egzaminoRezultatas)
00026             : Human(vardas, pavarde), pazymiai(pazymiai), egzaminoRezultatas(egzaminoRezultatas) {
00027             calculate_everything();
00028         }
00029
00030         // Destructor
00031         ~Studentas() {
00032             // cout << "Destruktorius iskviestas" << endl;
00033         }
00034
00035         // Copy constructor
00036         Studentas(const Studentas& other);
00037
00038         // Move constructor
00039         Studentas(Studentas&& other) noexcept;
00040
00041         // Copy constructor operator
00042         Studentas& operator=(const Studentas& other);
00043
00044         // Move constructor operator
00045         Studentas& operator=(Studentas&& other);
00046
00047
00048         // Setters
00049         void setVardas(const string& v) { vardas = v; }
00050         void setPavarde(const string& p) { pavarde = p; }
00051         void setGrades(const vector<float>& g) { pazymiai = g; }
00052         void setExam(int exam) { egzaminoRezultatas = exam; }
00053
00054         // Getters
00055         vector<float> getPazymiai() const { return pazymiai; }
00056         int getEgzaminoRezultatas() const { return egzaminoRezultatas; }
00057         float getGalutinisVid() const { return galutinisVid; }
00058         float getGalutinisMed() const { return galutinisMed; }
00059
00060
00061         // Member functions
00062         float calculateMean();
00063         float calculateMedian();
00064         void calculate_everything(); // Calls both calculateMean and calculateMedian and assigns them.
00065
00066
00067         // Friend function for printing
00068         friend ostream& operator<<(ostream &out, const Studentas &s);
00069
00070         // Friend function for input
00071         friend std::istream& operator>>(std::istream& in, Studentas& s);
00072

```

```
00073     // File functions
00074     static void appendingContainerViaFile(string file_name, vector <Studentas> &student_list);
00075     static vector<Studentas> read_student_records(int ndCount, std::istream& iss);
00076     static void print_to_file(vector<Studentas> list_of_students, string file_name);
00077 };
00078
```

5.9 src/additional_functions.cpp File Reference

```
#include "my_library.h"
#include "my_functions.h"
```

Functions

- int [check_the_value](#) (string message, string errorMessage, int minVal, int maxVal)
- string [consoleText_userChoice](#) ()

5.9.1 Function Documentation

5.9.1.1 check_the_value()

```
int check_the_value (
    string message,
    string errorMessage,
    int minVal,
    int maxVal)
```

5.9.1.2 consoleText_userChoice()

```
string consoleText_userChoice ()
```

5.10 src/file_functions.cpp File Reference

```
#include "my_library.h"
#include "my_functions.h"
```

Functions

- string [create_folder](#) (string folder_name)
[file_functions.cpp](#)
- void [generate_files](#) (int student_size)

5.10.1 Function Documentation

5.10.1.1 create_folder()

```
string create_folder (  
    string folder_name)
```

[file_functions.cpp](#)

5.10.1.2 generate_files()

```
void generate_files (  
    int student_size)
```

5.11 src/main.cpp File Reference

```
#include "my_library.h"  
#include "my_functions.h"
```

Functions

- int [main](#) ()

5.11.1 Function Documentation

5.11.1.1 main()

```
int main ()
```

5.12 src/program_choice.cpp File Reference

```
#include "my_library.h"  
#include "my_functions.h"
```

Functions

- [Studentas userChoice_2](#) ()
 [program_choice.cpp](#)
- [Studentas userChoice_3](#) ()

5.12.1 Function Documentation

5.12.1.1 userChoice_2()

```
Studentas userChoice_2 ()
```

[program_choice.cpp](#)

5.12.1.2 userChoice_3()

```
Studentas userChoice_3 ()
```

5.13 src/random_generating.cpp File Reference

```
#include "my_library.h"  
#include "my_functions.h"
```

Functions

- int [randomNumber](#) (int a, int b)
- vector< float > [random_grade](#) ()
- string [random_name](#) ()
- string [random_last_name](#) ()

5.13.1 Function Documentation

5.13.1.1 random_grade()

```
vector< float > random_grade ()
```

5.13.1.2 random_last_name()

```
string random_last_name ()
```

5.13.1.3 random_name()

```
string random_name ()
```

5.13.1.4 randomNumber()

```
int randomNumber (  
    int a,  
    int b)
```

5.14 src/student.cpp File Reference

```
#include "my_library.h"
#include "student.h"
#include "my_functions.h"
```

Functions

- ostream & operator<< (ostream &out, const [Studentas](#) &s)
- std::istream & operator>> (std::istream &in, [Studentas](#) &s)

5.14.1 Function Documentation

5.14.1.1 operator<<()

```
ostream & operator<< (
    ostream & out,
    const Studentas & s)
```

5.14.1.2 operator>>()

```
std::istream & operator>> (
    std::istream & in,
    Studentas & s)
```

5.15 src/student_functions.cpp File Reference

```
#include "my_library.h"
#include "my_functions.h"
```

Functions

- void [print_students](#) (vector< [Studentas](#) > &studentas)
student_functions.cpp
- void [insert_student](#) (vector< [Studentas](#) > &student_list, [Studentas](#) &student)
- void [sort_students](#) (vector< [Studentas](#) > &student_list)
- void [sort_students](#) (vector< [Studentas](#) > &student_list, int choice)

5.15.1 Function Documentation

5.15.1.1 insert_student()

```
void insert_student (
    vector< Studentas > & student_list,
    Studentas & student)
```

5.15.1.2 print_students()

```
void print_students (
    vector< Studentas > & studentas)
```

[student_functions.cpp](#)

5.15.1.3 sort_students() [1/2]

```
void sort_students (
    vector< Studentas > & student_list)
```

5.15.1.4 sort_students() [2/2]

```
void sort_students (
    vector< Studentas > & student_list,
    int choice)
```

5.16 src/testavimas.cpp File Reference

```
#include "my_library.h"
#include "my_functions.h"
```

Functions

- string [consoleText_testing12](#) ()
- template<typename Func>
double [measureTime](#) (Func func)
- void [testing_v12](#) ()
- void [testing_v11](#) ()
- vector< [Studentas](#) > [divide_students_unoptimized](#) (const vector< [Studentas](#) > &list_of_students, int choice)
- void [strategy_1](#) ()
- void [strategy_2](#) ()
- void [strategy_3](#) ()

5.16.1 Function Documentation

5.16.1.1 consoleText_testing12()

```
string consoleText_testing12 ()
```

5.16.1.2 divide_students_unoptimized()

```
vector< Studentas > divide_students_unoptimized (
    const vector< Studentas > & list_of_students,
    int choice)
```

5.16.1.3 measureTime()

```
template<typename Func>
double measureTime (
    Func func)
```

5.16.1.4 strategy_1()

```
void strategy_1 ()
```

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5.16.1.5 strategy_2()

```
void strategy_2 ()
```

5.16.1.6 strategy_3()

```
void strategy_3 ()
```

5.16.1.7 testing_v11()

```
void testing_v11 ()
```

5.16.1.8 testing_v12()

```
void testing_v12 ()
```


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